U-S-00156C(DSA-DM)
April 17, 1973
SUPERSEDING
Int. Fed. Spec.
U-S-00156b
March 16, 1970 and
Interim Revision of
Fed. Spec. U-S-156
October 8, 1948

INTERIM FEDERAL SPECIFICATION

SEALING COMPOUND, PULP CANAL, DENTAL

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification U-S-156.

1. SCOPE

1.1 Scope. This specification covers a sealing compound suitable for use as a pulp (root) canal sealer (cement) in endodontic procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-C-186	Container, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

FSC 6520

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplement as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Alanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, D.C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN PHARMACEUTICAL ASSOCIATION
National Formulary

(Application for copies should be addressed to the Mack Publishing Company, Easton Pa. 18042.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Powder. The sealing compound powder shall be of the following composition by weight, when tested as specified in 4.4.1:

Zinc Oxide, U.S.P., 1/ except arsenic not to exceed 0.0002 percent (%)	38.6 ± 3.0%
Hydrogenated resin, FDA 2/ grade (approved for use in contact with food), and in compliance with requirements for Rosin, N.F. 3/	28.8 ± 2.5%
Bismuth Subcarbonate, U.S.P.	14.4 ± 1.4%
Barium Sulfate, U.S.P.	14.4 ± 1.4%
Sodium Borate, U.S.P.	3.8 ± 1.0%

1/ Pharmacopeia of the United States of America
2/ Food and Drug Administration
3/ National Formulary

3.1.2 Liquid. The sealing compound liquid shall consist of Eugenol, U.S.P. The liquid shall be clear and free from deposits or sediment.

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3.2 Description. The sealing compound shall be of the non-staining type and shall be suitable for luting and hermetically sealing silver or gutta percha points in endodontically treated teeth. The sealing compound shall be non-toxic, non-irritating, radiopaque, water insoluble, and shall not discolor tooth structure.

3.3 Properties of mix. The sealing compound, when mixed according to manufacturer's instructions, shall exhibit the properties enumerated in 3.3.1 through 3.3.3.

3.3.1 Working time and appearance. The mixture shall produce a smooth, creamy, homogeneous paste of light cream color, when tested as specified in 4.4.2. The mixture shall have a working time of 15 to 20 minutes, when tested as specified in 4.4.2.

3.3.2 Hardening time. The mixture shall reach its final set, and shall have a tough, rigid body in 10 hours, when tested as specified in 4.4.3.

3.3.3 Adhesiveness and solubility. The mixture shall firmly adhere to a glass mixing slab and dissolve in chloroform, when tested as specified in 4.4.4.

3.4 Toxicity and irritation. The manufacturer shall furnish, at time of award, a statement directly to the Directorate of Medical Materiel (Code ATTS-2), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, that the sealing compound, when clinically tested and used according to manufacturer's instructions, has no toxic effect on a normal, healthy person, is non-irritating to periapical tissue, and provides a hermetical seal in the pulp canal.

3.5 Manufacturer's instructions. Adequate and accurate instructions for mixing and use of the material shall accompany each unit package. These instructions shall include the following as a minimum requirement:

(a) Powder/liquid ratio in terms of number of drops of liquid to grams of powder.

(b) Method of mixing including armamentaria necessary for use.

(c) Spatulating time required and its effect on setting time.

(d) Hardening time on a glass slab and setting time in the root canal.

Two copies of the instructions shall be forwarded to the contracting officer for information purposes.

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3.6 Immediate container. Immediate container of sealing compound shall consist of a bottle of powder and a bottle of liquid. In addition, the unit package shall include an instruction sheet and a glass dropper in a bakelite screw cap closure for the liquid. The powder and liquid shall each be furnished in properly sealed bottles suitable for containing the material without deterioration or contamination. A minimum of 14.2 grams of powder shall be furnished in a 1/2 ounce (oz.) wide-mouth light-resistant glass bottle with bakelite screw cap and shrink seal. A minimum of 7.5 cubic centimeters (c.c.) of liquid shall be furnished in a 1/4 fluid oz. narrow-mouth light-resistant glass bottle with bakelite screw cap and shrink seal. Immediate container labeling shall include lot or control number and the date of manufacture. The powder shall comply with the requirements for group A, class 1, type d or e, style 2, grade 1, closure A with outer seal of PPP-C-186. Sealer liquid shall comply with the requirements for group A, class 1, type d or e, style 1, grade 1, closure A (with outer seal) of PPP-C-00186a.

3.7 Stability. The sealing compound shall comply with the requirements of this specification for at least 36 months from date of shipment, when stored at 59 to 86 degrees Fahrenheit in a dark, dry place.

3.8 Delivery schedule. No more than six months shall have elapsed from the date of manufacture of the sealing compound to the date of delivery to the Government.

3.9 Lot or control number. Each unit package shall be marked with lot or control number, and date of manufacture.

3.10 Workmanship. Workmanship shall be first class throughout. The sealing compound shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the packaging and raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the material, the lot number, individual test results, and the specification requirements. In addition, when certificates of quality are furnished for raw materials, the manufacturer shall perform identification testing of the raw materials prior to formulation.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one package of sealing compound.

Table I. Acceptable Quality Levels

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with Table II. The acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one package of sealing compound. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The maximum size of a batch of powder shall be 50 pounds.

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Table II. Tests

Characteristic	Requirement	Test procedure	Minimum number of determinations per lot or batch number
Powder analysis	3.1.1	4.4.1	2
Working time and appearance	3.3.1	4.4.2	3
Hardening time	3.3.2	4.4.3	3
Adhesiveness and solubility	3.3.3	4.4.4	3
Net content of powder and liquids	3.6	S.M.I. 1/	5

1/ Standard measuring instruments.

4.3 Examination. The sealing compound shall be examined for defects including, but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Major:	
101	Unit of issue of sealing compound not complete with powder, liquid, instructions and dropper.
102	Powder not packaged in wide-mouthed, light-resistant glass bottles of 1/2 oz. size.
103	Sealer liquid not packaged in light-resistant glass bottles of 1/4 oz. size.
104	Cap is not tight.
105	Closure not of type specified.
106	Dropper cracked or broken.
107	Glass containers cracked or broken.
108	Liquid is turbid.
109	Liquid not free from deposits or sediment.
110	Powder not free from extraneous material such as dirt and lint.
111	Instructions not included.
112	Cap does not have nonabsorbent lining material.
113	Liquid leaks from closure.
Minor:	
201	Labeling not legible or not free from defacing marks and scratches.
202	Lining material does not remain in cap when cap is removed.
203	Packaging, packing and marking not as specified.

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4.4 Tests.

4.4.1 Powder assay. The methods of assay for each ingredient in the finished powder shall be submitted with the bid or proposal, and shall be subject to approval and acceptance by the Government. If the methods are approved and accepted by the Government, such methods shall be used to assay the ingredients. No further submission of methods will be required, providing the contractor states that the tests have not been modified in any way.

4.4.2 Working time and appearance. The sealing powder shall be spatulated with the sealing liquid, in accordance with manufacturer's instructions, on a clean, dry, glass mixing slab, for 4 minutes at 72 to 75 degrees Fahrenheit and at 40 to 45 percent relative humidity. At 30 second intervals, beginning 4-1/2 minutes from start of mix, the sealing compound shall be peaked using a suitable instrument. The instrument shall be wiped clean before each insertion. The time shall be recorded to the nearest 1/2 minute before the compound fails to slump to its own level. The test shall be repeated 3 times, and the sealing compound shall be left on the slab for further testing.

4.4.3 Hardening time. The test samples prepared in accordance with 4.4.2 shall be allowed to stand for 10 hours at constant temperature and humidity and shall be tested for hardness with a sharp instrument.

4.4.4 Adhesiveness and solubility. The test samples prepared in accordance with 4.4.3 shall adhere so firmly to the glass slab that they can be chipped off only with a knife or other sharp instrument. The test samples shall then be readily removable from the slab by the use of chloroform and a wiping cloth.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. One package consisting of one 1/2 oz. bottle containing a minimum of 14.2 gms. of powder, one 1/4 oz bottle containing a minimum of 7.5 cc of liquid, one separate dropper, and directions for use, as specified, constitute one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	96 units

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5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a box of appropriate size, constructed in accordance with PPP-B-676, type I, except that commercial colors are acceptable. Box shall be provided with a styrofoam fitting or equal so that each bottle and the dropper is securely held in an individual cell. Closure of box shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676 except that commercial colors are acceptable or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Fiberboard boxes shall conform to the special requirements specified in table I of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

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5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. Fiberboard boxes shall conform to the special requirements specified in table II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. In addition, the following legend shall be shown:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)"

The date of manufacture shall be shown in lieu of the date packed.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number and the following legend shall be shown:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)"

The date of manufacture shall be shown in lieu of the date packed.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification coverage. This document covers the requirements for FSN 6520-083-6587, SEALING COMPOUND, PULP CANAL, DENTAL, Radiopaque.

6.3 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1350

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

U-S-00156C (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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